

Introduction to Amazon Elastic Compute Cloud (EC2)

Lab Details

1. This lab walks you through the steps to launch and configure a virtual machine in the Amazon cloud.
2. You will practice using Amazon Machine Images to launch Amazon EC2 Instances and use key pairs for SSH authentication to log into your instance. You will create a web page and publish it.
3. Duration: **30 minutes**
4. AWS Region: **US East (N. Virginia) us-east-1**

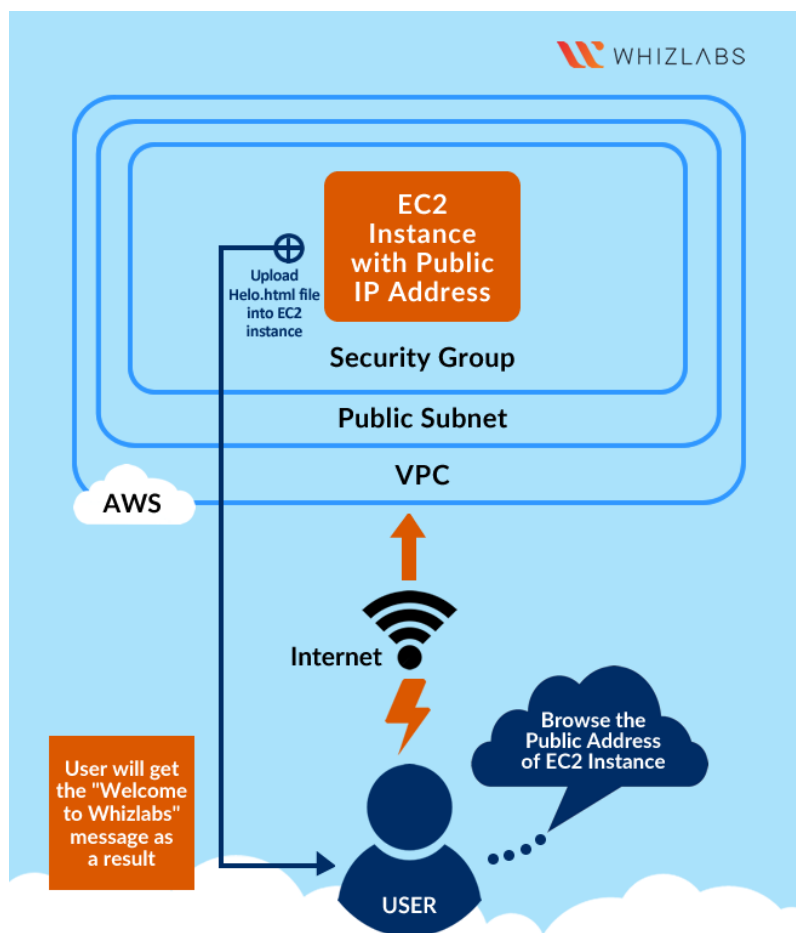
Introduction

What is EC2

- AWS defines it as Elastic Compute Cloud.
- It's a virtual environment where "you rent" to have your environment created, without purchasing.
- Amazon refers to these virtual machines as Instances.
- Preconfigured templates can be used to launch instances. These templates are referred to as images. Amazon provides these images in the form of AMIs (Amazon Machine Images).
- Allows you to install custom applications and services.
- Scaling of infrastructure i.e., up or down is easy based on the demand you face.
- AWS provides multiple configurations of CPU, memory, storage etc., through which you can pick the flavor that's required for your environment.
- No limitation on storage. You can pick the storage based on the type of the instance that you are working on.
- Temporary storage volumes are provided, which are called Instance Store Volumes. Data stored in this gets deleted once the instance is terminated.
- Persistent storage volumes are available and are referred to as EBS (Elastic Block Store) volumes.
- These instances can be placed at multiple locations which are referred to as Regions and Availability Zones (AZ).
- You can have your Instances distributed across multiple AZs i.e., within a single Region, so that if an instance fails, AWS automatically remaps the address to another AZ.

- Instances deployed in one AZ can be migrated to another AZ.
- To manage instances, images, and other EC2 resources, you can optionally assign your own metadata to each resource in the form of tags.
- A Tag is a label that you assign to an AWS resource. It contains a key and an optional value, both of which are defined by you.
- Each AWS account comes with a set of default limits on the resources on a per-Region basis.
- For any increase in the limit you need to contact AWS.
- To work with the created instances, we use Key Pairs.

Architecture Diagram



Task Details

1. Launching Lab Environment
2. Launching an EC2 Instance
3. SSH into EC2 Instance

4. Install an Apache Server
5. Create and publish page
6. Validation of the lab

Lab Steps

Task 1 : Launching an EC2 Instance

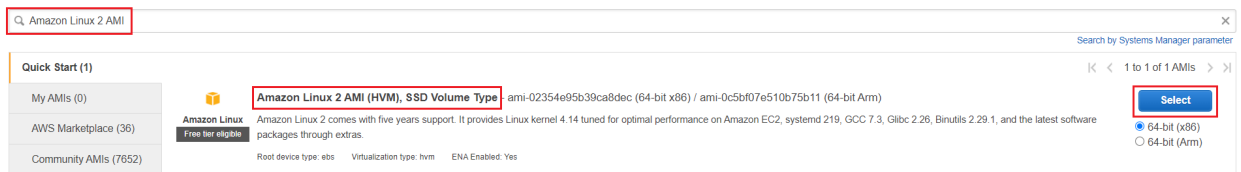
1. Make sure you are in **US East (N. Virginia) us-east-1** Region.
2. Navigate to EC2 by clicking on the **Services** menu in the top, then click on

EC2 in the  **Compute** section.

3. Navigate to **Instances** on the left panel and click on **Launch instances**.
4. **Choose an Amazon Machine Image (AMI):** Search for **Amazon Linux 2 AMI** in the search box and click on the **select** button.

Step 1: Choose an Amazon Machine Image (AMI)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace, or you can select one of your own AMIs.



5. **Choose an Instance Type:** select **t2.micro** and then click on the

Next: Configure Instance Details

6. **Configure Instance Details:** No need to change anything in this step, click on

Next: Add Storage

Note: Ignore the warning in the IAM Role selection section

8. **Add Storage:** No need to change anything in this step, click on

Next: Add Tags

9. **Add Tags:** Click on

Add Tag

- Key : Enter **Name**
- Value : Enter **MyEC2Server**

Next: Configure Security Group

- Click on

10. **Configure Security Group:**

- Assign a security group : Select **Create a new security group**
- Security group name : Enter **MyEC2Server_SG**
- Description : Enter **Security Group to allow traffic to EC2**
- To add **SSH**,

- Choose Type:
- Source: Select (From ALL IP addresses accessible).

- For **HTTP**,

- Click on
- Choose Type: **HTTP**
- Source: Select (From ALL IP addresses accessible).

- After that, click on

11. **Review and Launch :** Review all settings and click on .

12. **Key Pair :** Select **Create a new key pair**

- Key pair type : Select **RSA**
- Key pair name : Enter **MyEC2Key**

- click on , and store it on your local machine, Click on .

13. **Launch Status:** Your instance is now launching, Click on the instance ID and wait for complete initialization of the instance till status changes to **Running**.

Filter instances						
search: i-0e7c65791fd37c46b X		Clear filters				
☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐		i-0e7c65791fd37c46b	 Running	t2.micro	 Initializing	No alarms

8. Note down the sample IPv4 Public IP Address of the EC2 instance. A sample is shown in the screenshot below.

Details
Security
Networking
Storage
Status checks
Monitoring
Tags

▼ Instance summary Info

Instance ID

i-02470739fac726925

Public IPv4 address

34.207.74.4 open address

Task 2 : SSH into EC2 Instance

1. Select your EC2 instance(**MyEC2Server**) and click on the **Connect** button.

Instances (1/1) Info

Filter instances

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☒	MyEC2Server	i-08ab22dd9a79677bd	Running	t2.micro	2/2 checks passed	No alarms

2. Select **EC2 Instance Connect** option and click on **Connect** button.(Keep everything else as default)

Connect to instance [Info](#)
Connect to your instance i-08ab22dd9a79677bd (MyEC2Server) using any of these options

EC2 Instance Connect

Session Manager

SSH client

EC2 Serial Console

Instance ID
i-08ab22dd9a79677bd (MyEC2Server)

Public IP address
54.243.13.174

User name
ec2-user

Connect using a custom user name, or use the default user name ec2-user for the AMI used to launch the instance.

Note: In most cases, the guessed user name is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI user name.

Cancel

Connect

- A new tab will open in the browser where you can execute the CLI Commands.

```
< → ↺ console.aws.amazon.com/ec2/v2/connect/ec2-user/i-08ab22dd9a79677bd
Last login: Sat Sep 25 11:57:54 2021 from ec2-18-206-107-26.compute-1.amazonaws.com

  _|  _|_ )
 _| ( _|_ /  Amazon Linux 2 AMI
 _|\_|_|_|

https://aws.amazon.com/amazon-linux-2/
[ec2-user@ip-172-31-84-26 ~]$
```

Task 3: Install an Apache Server

- Switch to root user: `sudo su`
- Now run the updates using the following command:
 - `yum -y update`
- Once completed, lets install and run an apache server
 - Install the Apache web server:
 - `yum install httpd -y`

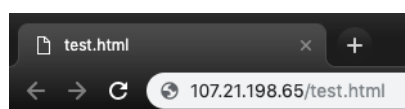
- Start the web server
 - `systemctl start httpd`
- Now enable httpd:
 - `systemctl enable httpd`
- Check the webserver status
 - `systemctl status httpd`
- You can see Active status is running.
- You can test that your web server is properly installed and started by entering the **public IPv4 address** of your **EC2 instance** in the address bar of a web browser. If your web server is running, then you see the Apache test page. If you don't see the Apache test page, then verify whether you followed the above steps properly and check your inbound rules for the security group that you created.
- Make sure the **URL Protocol** is **http** not https.

Syntax: `http://<Your_Public_IPv4_Address>`

Example: `http://34.207.74.4`

Task 4 : Create and publish the page

1. Navigate to the HTML folder where we will create an HTML page to test.
 - `cd /var/www/html/`
2. Create a sample **test.html** file using nano editor:
 - `nano test.html`
3. Enter sample HTML content provided below in the file and save the file with Ctrl+X and press "y" to confirm the save. Then press "Enter" to confirm the file name.
 - `<HTML> Hi Bamlabs, I am a public page </HTML>`
4. Restart the webserver by using the following command:
 - `systemctl restart httpd`
5. Now enter the file name, i.e., **/test.html** after the **public IPv4 Address** which you got when you created the ec2 instance in the browser, and you can see your HTML content.
 - Make sure **URL Protocol** is **http** not https.
 - Syntax: `http://<Your_Public_IPv4_Address>/test.html`
 - Sample URL: `http://107.21.198.65/test.html`



Hi Whizlabs, I am a public page

6. If you can see the above text in the browser, then you have successfully completed the lab.